

UNIT 1 – Introduction to Java Programming

No.	Question	Marks	RBT	Source Paper(s)
1	Explain JDK, JRE, JIT and JVM.	04	U	Summer-2026
2	Explain JVM in detail.	04	U	Summer-2024
3	Explain Structure of Java Program with suitable example.	04	U	Winter-2024
4	Explain Type Conversion and Casting in java.	03	U	Winter-2025
5	Explain the Java Program Structure with example.	03	U	Summer-2024 / Winter-2025
6	Explain the role of the compiler and interpreter in Java program execution.	04	U	Winter-2025
7	Illustrate different categories of operators in Java with code examples and explain how operator precedence affects output.	07	A	Winter-2025
8	List all Java features and explain any two of them in details.	03	U	Winter-2025
9	List arithmetic operators in Java. Develop a Java program using any three arithmetic operators and show the output of program.	07	A	Winter-2024
10	List different features of Java. Explain any four Features of Java.	04	U	Winter-2025
11	List out different types of operators in Java. Explain Arithmetic and Logical Operators in detail.	04	U	Summer-2025
12	List out features of JAVA and explain any two of them.	03	U	Summer-2026 / Winter-2025
13	List out various Primitive data types in Java.	03	R	Winter-2024
14	List relational operators in Java. Develop a Java program using any three relational operators and show the output of program.	07	A	Summer-2026
15	Write a Java program to demonstrate all types of type casting (implicit and explicit). Explain each conversion with output.	07	A	Winter-2025

UNIT 2 – Object-Oriented Programming Concepts

No.	Question	Marks	RBT	Source Paper(s)
1	Define Constructor. List types of Constructors. Develop a java code to explain Default constructor.	07	A	Summer-2026
2	Define Constructor. List types of Constructors. Develop a java code to explain Parameterized constructor.	07	A	Winter-2024
3	Define access modifiers. List the four types used in Java.	03	R	Winter-2025
4	Define: Constructor. List out types of it. Explain Parameterized and copy constructor with suitable example.	07	U	Winter-2025
5	Differentiate between Procedure-Oriented Programming (POP) and Object-Oriented Programming (OOP).	03	U	Winter-2025
6	Explain Garbage collection in java.	03	U	Summer-2024 / Winter-2025
7	Explain concept of static word and design a Java program that uses static blocks, static variables, and static methods.	07	A	Winter-2025
8	Explain different access controls in Java.	04	U	Summer-2024 / Winter-2025
9	Explain final keyword with example.	04	U	Summer-2024 / Winter-2024 / Summer-2026 / Winter-2025
10	Explain method overloading in JAVA with example.	04	U	Winter-2025
11	Explain static in java with example.	04	U	Summer-2026
12	Explain static keyword with example.	04	U	Summer-2024 /

				Winter-2024
13	Explain the role of the final keyword in variables and methods.	03	U	Winter-2025
14	Explain this in Java with example.	03	U	Summer-2026
15	Explain this keyword with suitable example.	03	U	Summer-2024
16	Give four Difference between Procedure Oriented Programming and Object-Oriented Programming.	04	U	Summer-2024 / Summer-2025
17	Give the basic syntax of the following terms with an example: (1) To create a Class, (2) To create an Object, (3) To define a Method, (4) To declare a Variable.	04	A	Summer-2025
18	List the basic OOP concepts in Java and explain any one.	03	U	Winter-2024
19	List the differences between Procedure-Oriented Programming (POP) and Object-Oriented Programming (OOP).	03	U	Winter-2024
20	What is constructor? Explain parameterized constructor with example.	07	U	Summer-2024
21	Write a program in Java that shows the use of Default Constructor.	03	A	Summer-2025
22	Write a program in Java that shows the use of parameterized Constructor.	03	A	Summer-2025
23	Write a program in Java which has a class Student having two instance variables enrollmentNo and name. Create 3 objects of Student class in main method and display student's name.	07	A	Summer-2025
24	Write a short note on constructor overloading with an example program.	04	A	Winter-2025
25	Write scope of java access modifier. Develop a java code to explain public modifier.	07	A	Winter-2024

UNIT 3 – Inheritance, Interfaces and Packages

No.	Question	Marks	RBT	Source Paper(s)
1	Compare 'this' keyword Vs 'Super' keyword. Explain super keyword with suitable Example.	07	U	Winter-2024
2	Define Inheritance. List out types of it. Explain multilevel and hierarchical inheritance with suitable example.	07	U	Summer-2024 / Winter-2025
3	Define Interface in Java. Write a Java program to demonstrate Multiple inheritance using interfaces in Java.	07	A	Summer-2026
4	Define abstract class in JAVA with example.	04	U	Summer-2025
5	Define an interface and mention how it differs from a class.	03	U	Winter-2025
6	Define package and write the syntax to create a package with example.	04	U	Summer-2025
7	Define package in JAVA. Write a Java program to create package.	03	A	Summer-2026
8	Discuss in detail the Object class methods (toString(), equals(), hashCode()) and show their overriding in a Java class Student.	07	A	Winter-2025
9	Explain Abstract class in Java with suitable example.	04	U	Summer-2026
10	Explain Final class in Java with suitable example.	04	U	Summer-2026
11	Explain Inheritance and its types in Java with suitable diagrams.	07	U	Summer-2026
12	Explain abstract class with suitable example.	03	U	Summer-2024
13	Explain method overriding with Java example.	04	U	Winter-2025
14	Explain package and list out steps to create package with suitable example.	07	U	Winter-2024
15	Explain super keyword with example.	04	U	Winter-2025
16	Explain use of packages in java with java programme.	04	A	Winter-2025
17	Give differences between Abstract class and Interface.	03	U	Winter-2024
18	List out any four differences between method overloading and overriding. Write a java code to explain method overriding.	07	A	Winter-2024
19	List out different types of inheritance and explain any one with example.	03	U	Winter-2024

20	What is interface? Explain multiple inheritance with example.	07	U	Winter-2025
21	What is package? Write steps to create a package and give example of it.	04	U	Summer-2024
22	Write a Java program to demonstrate single and multilevel inheritance using the extends keyword.	03	A	Winter-2025
23	Write a program in Java to demonstrate method overriding.	07	A	Summer-2025
24	Write a program in Java to demonstrate single inheritance.	03	A	Summer-2025 / Summer-2026
25	Write a program in Java to implement multiple inheritance using interfaces.	07	A	Summer-2025
26	Write a program to explain method overloading and method overriding.	07	A	Winter-2025

UNIT 4 – Exception Handling and Multithreading

No.	Question	Marks	RBT	Source Paper(s)
1	Define multi-threading and mention two of its advantages.	03	U	Winter-2025
2	Describe the complete lifecycle of a thread.	03	U	Summer-2025
3	Explain Array Index Out Of Bound Exception in Java with example.	03	U	Winter-2025
4	Explain Divide by Zero Exception in Java with example.	03	U	Winter-2025
5	Explain multiple catch with suitable example.	04	U	Winter-2025
6	Explain the purpose of the finally block in Java.	03	U	Winter-2025
7	Explain throw and finally in Exception Handling with example.	04	U	Summer-2024 / Winter-2025
8	Explain try catch block with example in Java.	03	U	Summer-2026
9	Explain try catch block with example.	04	U	Winter-2024
10	List and explain different types of errors in Java.	03	U	Summer-2025
11	List any four inbuilt exceptions in Java.	03	R	Summer-2024 / Summer-2026
12	List types of errors in Java.	03	R	Winter-2024
13	What is Exception? Write a program that shows the use of Arithmetic Exception.	07	A	Summer-2024 / Summer-2026
14	What is Thread? Explain Thread life cycle.	04	U	Summer-2024
15	What is thread ? Explain thread life cycle.	04	U	Summer-2025 / Summer-2026
16	Write a Java program that implements Runnable interface to perform two concurrent tasks (e.g., printing odd and even numbers).	07	A	Winter-2025
17	Write a Java program to demonstrate creation of threads by extending Thread class.	04	A	Winter-2025
18	Write a Java program to develop Banking Application in which user deposits the amount Rs 25000 and then start withdrawing of Rs 20000, Rs 4000 and it throws exception "Not Sufficient Fund" when user withdraws Rs. 2000 thereafter.	07	A	Summer-2025
19	Write a Java program to develop user defined exception for 'Divide by Zero' error.	07	A	Summer-2026
20	Write a Java program to develop user defined exception for "Divide by zero" error.	04	A	Winter-2024
21	Write a java program to explain finally block in exception handling.	04	A	Winter-2024
22	Write a program in java that creates the multiple threads by implementing the Thread class.	07	A	Summer-2024
23	Write a program that executes two threads. One thread displays "Thread1" every 1000 milliseconds, and the other displays "Thread2" every 2000 milliseconds. Create the threads by extending the Thread	07	A	Summer-2025

	class.			
24	Write a program to demonstrate multiple catch blocks for different exception types.	04	A	Winter-2025

UNIT 5 – Collections and File Handling

No.	Question	Marks	RBT	Source Paper(s)
1	Define serialization and state its purpose.	03	U	Winter-2025
2	Develop a program using ArrayList and LinkedList to store and display a list of student names. Compare both lists in terms of performance and usage.	07	A	Winter-2025
3	Differentiate between HashMap and TreeMap with suitable examples.	04	U	Winter-2025
4	Differentiate between String class and String Buffer class.	03	U	Winter-2025
5	Explain InputStream.	03	U	Summer-2024
6	Explain any two String buffer class methods with suitable example.	04	U	Winter-2024
7	Explain any two String class methods with suitable example.	04	U	Winter-2024
8	Explain the different I/O Classes available with Java.	04	U	Winter-2025
9	List Different Stream Classes in Java.	03	R	Summer-2026
10	List Different Stream Classes.	03	R	Summer-2024 / Winter-2024
11	List any three classes in the Collections Framework and their use.	03	R	Winter-2025
12	List different file operations in Java.	03	R	Summer-2026 / Summer-2024
13	What is stream class? How are the stream classes classified?	03	U	Summer-2025
14	Write a Java program to create and display ArrayList.	04	A	Summer-2026
15	Write a Java program to create and display LinkedList.	04	A	Summer-2026
16	Write a Java program to perform read and write operations on a Text file named Abc.txt.	07	A	Summer-2024
17	Write a Java program to read and write data from/to a text file using character streams.	04	A	Winter-2025
18	Write a java program to create a text file and perform read operation on the text file.	07	A	Summer-2024 / Summer-2026 / Winter-2025
19	Write a java program to create a text file and perform write operations on a Text File.	07	A	Winter-2025
20	Write a java program to display the content of a text file and perform append operation on the text file.	07	A	Summer-2024
21	Write a program in Java that reads the content of a file byte by byte and copy it into another file.	07	A	Winter-2024
22	Write a program in Java to create a file and perform write operation on this file.	07	A	Summer-2026
23	Write a program to demonstrate serialization and deserialization of a Student class object using ObjectOutputStream and ObjectInputStream.	07	A	Winter-2025